Week 9: Sales and Customer Analytics System

The digital retail enterprise requires deep data analytics across sales and customer transactional touchpoints to evaluate buyer behavior, assess product sales velocities, and map revenue trends. The executive leadership relies on aggregate reporting to highlight VIP customers, identify fast-moving inventory, and determine high-margin product categories.

1. Apply SQL Aggregate Functions

Aggregate functions perform calculations across multi-row records and produce a single consolidated summary value for reporting.

Functions Covered: Function	**Purpose**	**Example**
COUNT()	Counts number of records	Total customers/orders
SUM()	Calculates total value	Total sales amount
AVG()	Finds average value	Average order value
MIN()	Finds minimum value	Lowest product price
MAX()	Finds maximum value	Highest product price

2. Generate Total Sales Reports

Tasks:

Write SQL queries to calculate:

1. Total number of orders placed.

```
SQL> SELECT COUNT(*) AS NUMBER_OF_ORDERS_PLACED FROM ORDERS;

NUMBER_OF_ORDERS_PLACED
-----------------------
                     14
```

2. Total revenue generated from sales.

```
SQL> SELECT SUM(ORDER_TOTAL) AS TOTAL_REVENUE FROM ORDERS;

TOTAL_REVENUE
-------------
       206200
```

3. Average order value.

```
SQL> SELECT ROUND(AVG(ORDER_TOTAL), 2) AS AVERAGE_ORDER_VALUE FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
           14728.57
```

4. Highest and lowest order amount.

```
SQL> SELECT MAX(ORDER_TOTAL) AS HIGHEST_ORDER,
  2  MIN(ORDER_TOTAL) AS LOWEST_ORDER
  3  FROM ORDERS;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
        55850         2450
```

5. Total sales for a specific period.

```
SQL> SELECT SUM(ORDER_TOTAL) AS TOTAL_REVENUE FROM ORDERS
  2  WHERE ORDER_DATE BETWEEN DATE '2026-06-01' AND DATE '2026-08-31';

TOTAL_REVENUE
-------------
        37450
```

3. Customer Purchase Analysis

1. Total number of orders placed by each customer.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS TOTAL_ORDERS
  5  FROM CUSTOMERS C
  6  LEFT JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUST_ID  CUSTOMER_NAME                                        TOTAL_ORDERS
-------- -------------------------------------------------- ------------
CUST0001 Arjun Kumar                                                   1
CUST0015 Nithin Murugan                                                1
CUST0003 Rahul Menon                                                   1
CUST0004 Ananya Iyer                                                   1
CUST0005 Vikram Reddy                                                  1
CUST0006 Meera Krishnan                                                1
CUST0007 Suresh Nair                                                   1
CUST0008 Kavya Rajan                                                   1
CUST0009 Karthik Subramanian                                           1
CUST0010 Divya Pillai                                                  1
CUST0011 Aditya Verma                                                  1

CUST_ID  CUSTOMER_NAME                                        TOTAL_ORDERS
-------- -------------------------------------------------- ------------
CUST0012 Sneha Patel                                                   1
CUST0002 Priya Sharma                                                  1
CUST0014 Lakshmi Venkatesan                                            1
CUST0013 Rohan Bose                                                    0

15 rows selected.
```

2. Total amount spent by each customer.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       NVL(SUM(O.ORDER_TOTAL), 0) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMERS C
  6  LEFT JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUST_ID  CUSTOMER_NAME                                        TOTAL_AMOUNT_SPENT
-------- -------------------------------------------------- ------------------
CUST0001 Arjun Kumar                                                     55850
CUST0014 Lakshmi Venkatesan                                              28000
CUST0003 Rahul Menon                                                     28000
CUST0005 Vikram Reddy                                                    22000
CUST0009 Karthik Subramanian                                             18500
CUST0015 Nithin Murugan                                                   8350
CUST0004 Ananya Iyer                                                      8300
CUST0011 Aditya Verma                                                     7000
CUST0006 Meera Krishnan                                                   7000
CUST0007 Suresh Nair                                                      6500
CUST0008 Kavya Rajan                                                      5700

CUST_ID  CUSTOMER_NAME                                        TOTAL_AMOUNT_SPENT
-------- -------------------------------------------------- ------------------
CUST0010 Divya Pillai                                                     5200
CUST0002 Priya Sharma                                                     3350
CUST0012 Sneha Patel                                                      2450
CUST0013 Rohan Bose                                                          0

15 rows selected.
```

NVL() function replaces NULL with a specified value. Here for the customer with no purchase the Amount Spent will be zero.

3. Average spending per customer.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       ROUND(NVL(AVG(O.ORDER_TOTAL), 0), 2) AS AVERAGE_SPENDING
  5  FROM CUSTOMERS C
  6  LEFT JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY AVERAGE_SPENDING DESC;

CUST_ID  CUSTOMER_NAME                                    AVERAGE_SPENDING
-------- ------------------------------------------------ ----------------
CUST0001 Arjun Kumar                                             55850
CUST0014 Lakshmi Venkatesan                                      28000
CUST0003 Rahul Menon                                             28000
CUST0005 Vikram Reddy                                            22000
CUST0009 Karthik Subramanian                                     18500
CUST0015 Nithin Murugan                                           8350
CUST0004 Ananya Iyer                                              8300
CUST0011 Aditya Verma                                             7000
CUST0006 Meera Krishnan                                           7000
CUST0007 Suresh Nair                                              6500
CUST0008 Kavya Rajan                                              5700

CUST_ID  CUSTOMER_NAME                                    AVERAGE_SPENDING
-------- ------------------------------------------------ ----------------
CUST0010 Divya Pillai                                             5200
CUST0002 Priya Sharma                                             3350
CUST0012 Sneha Patel                                              2450
CUST0013 Rohan Bose                                                  0

15 rows selected.
```

4. Customers with maximum purchase amount.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       SUM(O.ORDER_TOTAL) AS TOTAL_PURCHASE_AMOUNT
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  HAVING SUM(O.ORDER_TOTAL) = (
 13       SELECT MAX(TOTAL_SPENDING)
 14       FROM (
 15            SELECT SUM(ORDER_TOTAL) AS TOTAL_SPENDING
 16            FROM ORDERS
 17            GROUP BY CUST_ID
 18       )
 19  );

CUST_ID  CUSTOMER_NAME                                  TOTAL_PURCHASE_AMOUNT
-------- ------------------------------------------------ --------------------
CUST0001 Arjun Kumar                                               55850
```

5. Customers with fewer purchases.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMERS C
  6  LEFT JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  HAVING COUNT(O.ORDER_REF) <= 1
 13  ORDER BY NUMBER_OF_ORDERS;

CUST_ID  CUSTOMER_NAME                                    NUMBER_OF_ORDERS
-------- ------------------------------------------------ ----------------
CUST0013 Rohan Bose                                                      0
CUST0015 Nithin Murugan                                                  1
CUST0003 Rahul Menon                                                     1
CUST0004 Ananya Iyer                                                     1
CUST0005 Vikram Reddy                                                    1
CUST0006 Meera Krishnan                                                  1
CUST0007 Suresh Nair                                                     1
CUST0008 Kavya Rajan                                                     1
CUST0009 Karthik Subramanian                                             1
CUST0010 Divya Pillai                                                    1
CUST0011 Aditya Verma                                                    1

CUST_ID  CUSTOMER_NAME                                    NUMBER_OF_ORDERS
-------- ------------------------------------------------ ----------------
CUST0012 Sneha Patel                                                     1
CUST0014 Lakshmi Venkatesan                                              1
CUST0001 Arjun Kumar                                                     1
CUST0002 Priya Sharma                                                    1

15 rows selected.
```

4. Find Top Customers Based on Purchase Amount

1. Top 5 customers based on total spending.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       SUM(O.ORDER_TOTAL) AS TOTAL_SPENDING
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC
 13  FETCH FIRST 5 ROWS ONLY;

CUST_ID  CUSTOMER_NAME                                  TOTAL_SPENDING
-------- ------------------------------------------------ --------------
CUST0001 Arjun Kumar                                           55850
CUST0003 Rahul Menon                                           28000
CUST0014 Lakshmi Venkatesan                                    28000
CUST0005 Vikram Reddy                                          22000
CUST0009 Karthik Subramanian                                   18500
```

2. Customers with maximum number of orders.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  HAVING COUNT(O.ORDER_REF) = (
 13       SELECT MAX(ORDER_COUNT)
 14       FROM (
 15            SELECT COUNT(*) AS ORDER_COUNT
 16            FROM ORDERS
 17            GROUP BY CUST_ID
 18       )
 19  )
 20  ORDER BY CUSTOMER_NAME;

CUST_ID  CUSTOMER_NAME                                      NUMBER_OF_ORDERS
-------- -------------------------------------------------- ----------------
CUST0011 Aditya Verma                                                      1
CUST0004 Ananya Iyer                                                       1
CUST0001 Arjun Kumar                                                       1
CUST0010 Divya Pillai                                                      1
CUST0009 Karthik Subramanian                                               1
CUST0008 Kavya Rajan                                                       1
CUST0014 Lakshmi Venkatesan                                                1
CUST0006 Meera Krishnan                                                    1
CUST0015 Nithin Murugan                                                    1
CUST0002 Priya Sharma                                                      1
CUST0003 Rahul Menon                                                       1

CUST_ID  CUSTOMER_NAME                                      NUMBER_OF_ORDERS
-------- -------------------------------------------------- ----------------
CUST0012 Sneha Patel                                                       1
CUST0007 Suresh Nair                                                       1
CUST0005 Vikram Reddy                                                      1

14 rows selected.
```

3. Frequent customers.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMERS C
  6  JOIN ORDERS O
  7    ON C.CUST_ID = O.CUST_ID
  8  GROUP BY
  9       C.CUST_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  HAVING COUNT(O.ORDER_REF) >= 2
 13  ORDER BY NUMBER_OF_ORDERS DESC;

no rows selected
```

4. High-value customers.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS,
  5       SUM(O.ORDER_TOTAL) AS TOTAL_SPENDING
  6  FROM CUSTOMERS C
  7  JOIN ORDERS O
  8    ON C.CUST_ID = O.CUST_ID
  9  GROUP BY
 10       C.CUST_ID,
 11       C.FIRST_NAME,
 12       C.LAST_NAME
 13  HAVING SUM(O.ORDER_TOTAL) > 20000
 14  ORDER BY TOTAL_SPENDING DESC;

CUST_ID  CUSTOMER_NAME                                        NUMBER_OF_ORDERS
-------- -------------------------------------------------- ----------------
TOTAL_SPENDING
--------------
CUST0001 Arjun Kumar                                                       1
         55850

CUST0014 Lakshmi Venkatesan                                                1
         28000

CUST0003 Rahul Menon                                                       1
         28000

CUST_ID  CUSTOMER_NAME                                        NUMBER_OF_ORDERS
-------- -------------------------------------------------- ----------------
TOTAL_SPENDING
--------------
CUST0005 Vikram Reddy                                                      1
         22000
```

5. Identify Best-Selling Products

1. Products with maximum sales quantity.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       SUM(OD.ITEM_QTY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCTS P
  6  JOIN ORDER_ITEMS OD
  7    ON P.PROD_ID = OD.PROD_ID
  8  GROUP BY
  9       P.PROD_ID,
 10       P.ITEM_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PROD_ID ITEM_NAME                                          TOTAL_QUANTITY_SOLD
------- -------------------------------------------------- -------------------
P000006 USB Cable                                                            6
P000005 Smartphone                                                           6
P000015 Wireless Earbuds                                                     3
P000008 Smart Watch                                                          2
P000012 External Hard Disk                                                   2
P000009 Headphones                                                           2
P000001 Laptop                                                               1
P000013 Gaming Monitor                                                       1
P000010 Tablet                                                               1
P000014 Laptop Stand                                                         1
P000002 Wireless Mouse                                                       1

PROD_ID ITEM_NAME                                          TOTAL_QUANTITY_SOLD
------- -------------------------------------------------- -------------------
P000003 Mechanical Keyboard                                                  1
P000004 Bluetooth Speaker                                                    1

13 rows selected.
```

2. Products generating highest revenue.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       SUM(OD.ITEM_QTY) AS QUANTITY_SOLD,
  5       SUM(OD.ITEM_QTY * OD.UNIT_PRICE) AS TOTAL_REVENUE
  6  FROM PRODUCTS P
  7  JOIN ORDER_ITEMS OD
  8    ON P.PROD_ID = OD.PROD_ID
  9  GROUP BY
 10       P.PROD_ID,
 11       P.ITEM_NAME
 12  ORDER BY TOTAL_REVENUE DESC;

PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000005 Smartphone                                                     6
       168000

P000001 Laptop                                                         1
        55000

P000010 Tablet                                                         1
        22000
```

```
PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000013 Gaming Monitor                                                 1
        18500

P000008 Smart Watch                                                    2
        13000

P000015 Wireless Earbuds                                               3
        10500

PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000012 External Hard Disk                                             2
        10400

P000009 Headphones                                                     2
         8400

P000004 Bluetooth Speaker                                              1
         3200

PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000006 USB Cable                                                      6
         2700

P000003 Mechanical Keyboard                                            1
         2500

P000014 Laptop Stand                                                   1
         1500

PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000002 Wireless Mouse                                                 1
          850

13 rows selected.
```

3. Least-selling products.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       NVL(SUM(OD.ITEM_QTY), 0) AS QUANTITY_SOLD
  5  FROM PRODUCTS P
  6  LEFT JOIN ORDER_ITEMS OD
  7    ON P.PROD_ID = OD.PROD_ID
  8  GROUP BY
  9       P.PROD_ID,
 10       P.ITEM_NAME
 11  ORDER BY QUANTITY_SOLD ASC;

PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
P000007 Power Bank                                                     0
P000011 Web Camera                                                     0
P000004 Bluetooth Speaker                                              1
P000003 Mechanical Keyboard                                            1
P000001 Laptop                                                         1
P000002 Wireless Mouse                                                 1
P000014 Laptop Stand                                                   1
P000013 Gaming Monitor                                                 1
P000010 Tablet                                                         1
P000012 External Hard Disk                                             2
P000009 Headphones                                                     2

PROD_ID ITEM_NAME                                          QUANTITY_SOLD
------- -------------------------------------------------- -------------
P000008 Smart Watch                                                    2
P000015 Wireless Earbuds                                               3
P000005 Smartphone                                                     6
P000006 USB Cable                                                      6

15 rows selected.
```

4. Products requiring promotion.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       P.UNIT_PRICE,
  5       P.QTY_IN_STOCK,
  6       NVL(SUM(OD.ITEM_QTY), 0) AS QUANTITY_SOLD
  7  FROM PRODUCTS P
  8  LEFT JOIN ORDER_ITEMS OD
  9    ON P.PROD_ID = OD.PROD_ID
 10  GROUP BY
 11       P.PROD_ID,
 12       P.ITEM_NAME,
 13       P.UNIT_PRICE,
 14       P.QTY_IN_STOCK
 15  HAVING NVL(SUM(OD.ITEM_QTY), 0) <= 1
 16     AND P.QTY_IN_STOCK > 0
 17  ORDER BY QUANTITY_SOLD, P.ITEM_NAME;

PROD_ID ITEM_NAME                                          UNIT_PRICE
------- -------------------------------------------------- ----------
QTY_IN_STOCK QUANTITY_SOLD
------------ -------------
P000007 Power Bank                                             1800
          75             0

P000011 Web Camera                                             2800
          50             0

P000004 Bluetooth Speaker                                      3200
          45             1

PROD_ID ITEM_NAME                                          UNIT_PRICE
------- -------------------------------------------------- ----------
QTY_IN_STOCK QUANTITY_SOLD
------------ -------------
P000013 Gaming Monitor                                        18500
          20             1

P000001 Laptop                                                55000
          25             1

P000014 Laptop Stand                                           1500
          80             1

PROD_ID ITEM_NAME                                          UNIT_PRICE
------- -------------------------------------------------- ----------
QTY_IN_STOCK QUANTITY_SOLD
------------ -------------
P000003 Mechanical Keyboard                                    2500
          60             1

P000010 Tablet                                                22000
          30             1

P000002 Wireless Mouse                                          850
         100             1

9 rows selected.
```

6. Perform Category-wise Sales Analysis

1. Total sales generated by each category.

```
SQL> SELECT
  2       C.CAT_NAME,
  3       SUM(OD.ITEM_QTY * OD.UNIT_PRICE) AS TOTAL_SALES
  4  FROM CATEGORIES C
  5  JOIN PRODUCTS P
  6    ON C.CAT_REF = P.CAT_REF
  7  JOIN ORDER_ITEMS OD
  8    ON P.PROD_ID = OD.PROD_ID
  9  GROUP BY
 10       C.CAT_REF,
 11       C.CAT_NAME
 12  ORDER BY TOTAL_SALES DESC;

CAT_NAME                                           TOTAL_SALES
-------------------------------------------------- -----------
Mobile Devices                                          190000
Laptops                                                  55000
Audio Devices                                            22100
Monitors                                                 18500
Wearables                                                13000
Storage Devices                                          10400
Computer Accessories                                      3350
Power Accessories                                         2700
Laptop Accessories                                        1500

9 rows selected.
```

2. Number of products sold in each category.

```
SQL> SELECT
  2       C.CAT_NAME,
  3       SUM(OD.ITEM_QTY) AS PRODUCTS_SOLD
  4  FROM CATEGORIES C
  5  JOIN PRODUCTS P
  6    ON C.CAT_REF = P.CAT_REF
  7  JOIN ORDER_ITEMS OD
  8    ON P.PROD_ID = OD.PROD_ID
  9  GROUP BY
 10       C.CAT_REF,
 11       C.CAT_NAME
 12  ORDER BY PRODUCTS_SOLD DESC;

CAT_NAME                                           PRODUCTS_SOLD
-------------------------------------------------- -------------
Mobile Devices                                                 7
Power Accessories                                              6
Audio Devices                                                  6
Computer Accessories                                           2
Wearables                                                      2
Storage Devices                                                2
Laptop Accessories                                             1
Laptops                                                        1
Monitors                                                       1

9 rows selected.
```

3. Highest revenue-generating category.

```
SQL> SELECT
  2       C.CAT_NAME,
  3       SUM(OD.ITEM_QTY * OD.UNIT_PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORIES C
  5  JOIN PRODUCTS P
  6    ON C.CAT_REF = P.CAT_REF
  7  JOIN ORDER_ITEMS OD
  8    ON P.PROD_ID = OD.PROD_ID
  9  GROUP BY
 10       C.CAT_REF,
 11       C.CAT_NAME
 12  HAVING SUM(OD.ITEM_QTY * OD.UNIT_PRICE) = (
 13       SELECT MAX(CATEGORY_REVENUE)
 14       FROM (
 15            SELECT SUM(OD2.ITEM_QTY * OD2.UNIT_PRICE) AS CATEGORY_REVENUE
 16            FROM CATEGORIES C2
 17            JOIN PRODUCTS P2
 18              ON C2.CAT_REF = P2.CAT_REF
 19            JOIN ORDER_ITEMS OD2
 20              ON P2.PROD_ID = OD2.PROD_ID
 21            GROUP BY C2.CAT_REF
 22       )
 23  );

CAT_NAME                                           TOTAL_REVENUE
-------------------------------------------------- -------------
Mobile Devices                                          190000
```

4. Average sales per category.

```
SQL> SELECT
  2       C.CAT_NAME,
  3       ROUND(AVG(OD.ITEM_QTY * OD.UNIT_PRICE), 2) AS AVERAGE_SALES
  4  FROM CATEGORIES C
  5  JOIN PRODUCTS P
  6    ON C.CAT_REF = P.CAT_REF
  7  JOIN ORDER_ITEMS OD
  8    ON P.PROD_ID = OD.PROD_ID
  9  GROUP BY
 10       C.CAT_REF,
 11       C.CAT_NAME
 12  ORDER BY AVERAGE_SALES DESC;

CAT_NAME                                           AVERAGE_SALES
-------------------------------------------------- -------------
Mobile Devices                                             95000
Laptops                                                    55000
Monitors                                                   18500
Wearables                                                   6500
Storage Devices                                             5200
Audio Devices                                            3683.33
Computer Accessories                                        1675
Laptop Accessories                                          1500
Power Accessories                                            900

9 rows selected.
```

7. Generate Business Analytics Reports

Report 1: Sales Performance Report

- Total sales
- Number of orders
- Average order value

- Highest order value

```
SQL> SELECT
  2       SUM(ORDER_TOTAL) AS TOTAL_SALES,
  3       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS,
  4       ROUND(AVG(ORDER_TOTAL), 2) AS AVERAGE_ORDER_VALUE,
  5       MAX(ORDER_TOTAL) AS HIGHEST_ORDER_VALUE
  6  FROM ORDERS O;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- --------------- ------------------- -------------------
     206200              14            14728.57               55850
```

Report 2: Customer Analytics Report

- Customer name
- Number of orders
- Total spending
- Purchase frequency

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       COUNT(O.ORDER_REF) AS NUMBER_OF_ORDERS,
  4       NVL(SUM(O.ORDER_TOTAL), 0) AS TOTAL_SPENDING,
  5       COUNT(O.ORDER_REF) AS PURCHASE_FREQUENCY
  6  FROM CUSTOMERS C
  7  LEFT JOIN ORDERS O
  8    ON C.CUST_ID = O.CUST_ID
  9  GROUP BY
 10       C.CUST_ID,
 11       C.FIRST_NAME,
 12       C.LAST_NAME
 13  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
------------------------------ ---------------- --------------
PURCHASE_FREQUENCY
------------------
Arjun Kumar                                    1          55850
                 1

Lakshmi Venkatesan                             1          28000
                 1

Rahul Menon                                    1          28000
                 1

CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
------------------------------ ---------------- --------------
PURCHASE_FREQUENCY
------------------
Vikram Reddy                                   1          22000
                 1

Karthik Subramanian                            1          18500
                 1

Nithin Murugan                                 1           8350
                 1
```

```
CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
------------------------------ ---------------- --------------
PURCHASE_FREQUENCY
------------------
Ananya Iyer                                    1           8300
                 1

Aditya Verma                                   1           7000
                 1

Meera Krishnan                                 1           7000
                 1

CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
------------------------------ ---------------- --------------
PURCHASE_FREQUENCY
------------------
Suresh Nair                                    1           6500
                 1

Kavya Rajan                                    1           5700
                 1

Divya Pillai                                   1           5200
                 1

CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
------------------------------ ---------------- --------------
PURCHASE_FREQUENCY
------------------
Priya Sharma                                   1           3350
                 1

Sneha Patel                                    1           2450
                 1

Rohan Bose                                     0              0
                 0

15 rows selected.
```

Report 3: Product Performance Report

- Product name
- Quantity sold
- Revenue generated

```
SQL> SELECT
  2       P.ITEM_NAME,
  3       SUM(OD.ITEM_QTY) AS QUANTITY_SOLD,
  4       SUM(OD.ITEM_QTY * OD.UNIT_PRICE) AS REVENUE_GENERATED
  5  FROM PRODUCTS P
  6  JOIN ORDER_ITEMS OD
  7    ON P.PROD_ID = OD.PROD_ID
  8  GROUP BY
  9       P.PROD_ID,
 10       P.ITEM_NAME
 11  ORDER BY REVENUE_GENERATED DESC;
```

```
ITEM_NAME                                          QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Smartphone                                                     6
            168000

Laptop                                                         1
             55000

Tablet                                                         1
             22000

ITEM_NAME                                          QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Gaming Monitor                                                 1
             18500

Smart Watch                                                    2
             13000

Wireless Earbuds                                               3
             10500

ITEM_NAME                                          QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
External Hard Disk                                             2
             10400

Headphones                                                     2
              8400

Bluetooth Speaker                                              1
              3200

ITEM_NAME                                          QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
USB Cable                                                      6
              2700

Mechanical Keyboard                                            1
              2500

Laptop Stand                                                   1
              1500

ITEM_NAME                                          QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Wireless Mouse                                                 1
               850

13 rows selected.
```

Report 4: Category Analysis Report

- Category name
- Total products sold
- Total revenue

```
SQL> SELECT
  2       C.CAT_NAME,
  3       SUM(OD.ITEM_QTY) AS TOTAL_PRODUCTS_SOLD,
  4       SUM(OD.ITEM_QTY * OD.UNIT_PRICE) AS TOTAL_REVENUE
  5  FROM CATEGORIES C
  6  JOIN PRODUCTS P
  7    ON C.CAT_REF = P.CAT_REF
  8  JOIN ORDER_ITEMS OD
  9    ON P.PROD_ID = OD.PROD_ID
 10  GROUP BY
 11       C.CAT_REF,
 12       C.CAT_NAME
 13  ORDER BY TOTAL_REVENUE DESC;

CAT_NAME                                           TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
-------------------------------------------------- ------------------- -------------
Mobile Devices                                                       7        190000
Laptops                                                              1         55000
Audio Devices                                                        6         22100
Monitors                                                             1         18500
Wearables                                                            2         13000
Storage Devices                                                      2         10400
Computer Accessories                                                 2          3350
Power Accessories                                                    6          2700
Laptop Accessories                                                   1          1500

9 rows selected.
```